BAF: AN AUDIO FINGERPRINTING DATASET FOR BROADCAST MONITORING

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ABSTRACT

Audio Fingerprinting (AFP) is a well-studied problem in music information retrieval for various use-cases e.g. content-based copy detection, DJ-set monitoring, and music excerpt identification. However, AFP for continuous broadcast monitoring (e.g. for TV & Radio), where music is often in the background, has not received much attention despite its importance to the music industry. In this paper (1) we present BAF, the first public dataset for music monitoring in broadcast. It contains 74 hours of production music from Epidemic Sound and 57 hours of TV audio recordings. Furthermore, BAF provides cross-annotations with exact matching timestamps between Epidemic tracks and TV recordings. Approximately, 80% of the total annotated time is background music. (2) We benchmark BAF with public state-of-the-art AFP systems, together with our proposed baseline *PeakFP*: a simple, non-scalable AFP algorithm based on spectral peak matching. In this benchmark, none of the algorithms obtain a F1-score above 47%, pointing out that further research is needed to reach the AFP performance levels in other studied use cases. The dataset, baseline, and benchmark framework are open and available for research.

1. INTRODUCTION

Audio Fingerprinting (AFP) is the information retrieval task of identifying audio recordings in a given database of reference songs. The task is based on extracting content-based signatures that summarize an audio recording (*extraction*) [1], storing them in a database or in hash tables (*indexing*), and efficiently linking short snippets of unlabeled audio to the same content in the database (*matching*).

AFP has been successfully applied to different tasks such as query by example [2], advertisement tracking [3], integrity verification [4], and data deduplication [5]. A relevant AFP application is broadcast monitoring for its crucial role in royalties distribution. In 2021, US$ 2.9 billion were distributed among rights holders [6], representing 11.5% of the global recorded music industry revenues.

A great AFP system for broadcast monitoring must provide exact start and end timestamps within long audio recordings. It also must be robust against common distortions in broadcasting like music being in the background and with low SNR. To the best of our knowledge, the literature has not addressed the AFP use case of broadcasting monitoring with a heavy presence of background music. In this scenario, music may have a very low Signal-to-Noise ratio (SNR) that makes it difficult to identify [7]. Moreover, music may be masked by a large variety of non-musical sounds, such as speech, applause, laughter, urban and nature sounds, etc. [8].

To promote research in AFP for broadcast monitoring we propose BAF: a Broadcast Audio Fingerprinting dataset with TV recordings in which production music is played. The references are part of Epidemic Sound's private catalog [9], a collection of music for content creators (production music). BAF reflects a challenging (but realistic) scenario [7] for broadcast AFP systems: low sample rate monaural audio, background music of variable SNR, a large variety of contexts, long queries with true negative sections, and matches of multiple durations. More details about the dataset can be found in Section §3.

In order to show the challenges that BAF presents, in Section §4 we present a benchmark with 4 of the available AFP algorithms: Audfprint [10], Panako [11, 12], Olaf [13], and NeuralFP [14]. These are open-source implementations that represent different approaches to AFP. In addition, we propose an open implementation of a simple baseline based on spectral peak matching and the evaluation framework used in the benchmark.

2. RELATED DATASETS

In this section, we present public datasets that have been used in the AFP literature. We also gather information about the contents of private datasets and depict them in Table 1. We include information about the queries, references, and the goal of the dataset or publication. We have found in the literature 21 datasets that have been used for

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Work	(Synthetic) Queries	References	Goal	Used in
2002 Haitsma, J. [15]	(✓) 4 excerpts	20,000	Scalable, robust to noise	
2003 Wang, A. [2]	250 excerpts	10,000	Scalable, robust to noise	
2005 Bartsch, M.A. [16]	93	93	Structural redundancy	
2008 Baluja, S. [17]	(✓) 1,000	10,000	Robust to pitch and time	
2008 Bellettini, C. [18, 19]	(✓) 15,000	15,000	Robust to pitch	
2011 Fenet, S. [20]	7d radio broad.	7,309 (60s)	Scalable, robust to pitch	
2011 Fenet, S. [20]	5d radio broad.	30,000	Scalable, robust to pitch	
2014 Malekesmaeili, M. [21]	(✓) 200	200	Robust to noise, pitch and time	
2015 Zhang, X. [22]	(✓) 10s excerpts	2,075	Robust to pitch and time	
2016 Sonnleitner, R. [23]	(✓) 300	20,000	Robust to noise, pitch and time	
2016 Sonnleitner, R [24]	8 DJ-mixes	296 $\approx$ 7h	Robust to pitch and time	
2016 Walter, T. [25]	10s excerpts	10M	Efficient and scalable AFP	
2017 Gfeller, B. [26]	12,000 excerpts	450h	Low-power music recognizer	
2020 Son H.-S. [27]	(✓) 100	100	Robust to pitch	
2020 Yu, Z. [28]	(✓) 5,000 (10s)	345,000	Robust to any degradation	
2009 MagnaTagATune [29]		25,863 $\approx$ 208h	Music Tagging	[30]
2006 TRECVID [31, 32]	(✓) 201	11,200 $\approx$ 400h	Content-based Copy Detection	[33–35]
2014 Panako [11]	(✓) 600 excerpts	30,000 $\approx$ 277h	Robust to pitch and time	[11]
2016 Mixotic [24]	10 DJ-mixes	723 $\approx$ 11h	Robust to pitch and time	[24]
2016 QuadFP [23]	(✓) 450,000	100,011 $\approx$ 6,899h	Robust to pitch and time	[23]
2021 NeuralFP [14]	(✓) short excerpts	100k $\approx$ 8,000h	High-specific audio retrieval	[14]

Table 1. Private (top ↑) and public (bottom ↓) datasets that have been used for AFP. For each dataset information about the queries, references and the goal of the original work is given. Synthetic (✓) queries have been created by applying transformations to reference audios. We also indicate if the queries are excerpts and the number of original audio pieces that have been transformed, not meaning the total number of queries after the transformation.

AFP. Even though the nature of the data varies from each dataset, most of them rely on the same principle: a private collection of tracks that constitute a reference set and a query set formed by applying transformations to some of the references. This is a good way to test the limits of the robustness to degradations like pitch-shifting or time-scaling. Still, it does not target the characteristics of realistic broadcast monitoring, where the music is in the background, masked by speech and a wide variety of sounds and noises (See Section §3.3.1 for a detailed analysis).

As Table 1 reflects, only 6 out of the 21 datasets are public, mainly due to the difficulty of legally publishing copyrighted music. In other cases, data remains private to protect intellectual property, as with private companies. To that extent, private datasets hinder reproducibility and slow-down scientific progress in AFP research. Of all private datasets, only the ones built by Fenet et al. [20] reflect the use case of broadcast monitoring: they use real radio broadcasted emissions as queries and a reference set of 459 songs. Moreover, only Fenet's [20], Sonnleitner et al., [24] and Walter and Gould [25] used unknown audios as queries. All other works used a version of the reference songs often modified with some degradation: echo, pitch and/or tempo alteration, reverb, etc. Besides these private datasets, there are some other public datasets that have been used in AFP works, even though none of them fits the broadcast monitoring task.

MagnaTagATune [29] is a public dataset created for Music Tagging. Each audio clip has associated a vector of binary annotations of 188 tags that describe the music piece. The dataset was used for AFP by Ramona and Peeters [30] in which they followed the experimental protocol of Haitsma and Kalker [15] applying a series of distortions like Amplitude dynamic compression, MP3 encoding, time-shifting, or equalization to 500 music clips from MagnaTagATune.

NIST-TRECVID [31, 32]. One of the tasks the TREC Video Retrieval Evaluation (TRECVID) proposed until 2011 was Content-Based Copy Detection (CCD) [36]. Various works [33–35] used the dataset provided with the task to evaluate AFP algorithms. The length of queries varies from 3 to 180 seconds and compromises multiple transformed fragments from 201 unique audio recordings.

Mixotic [24] was created by Sonnleitner et al. to test the robustness of QuadFP, Panako, and Audfprint in real DJ mixes. It was generated from free, CC-licensed DJ mixes that were published on mixotic netlabel.

Panako [11], **QuadFP** [23, 37] and **NeuralFP** [14] present public datasets that were curated to test their algorithms. Since these datasets are reproducible we list them as public AFP datasets, but in the case of Panako and QuadFP they share a script that downloads tracks from Jamendo instead of the audio files. It can happen that some audio works are not available anymore thus impeding the reconstruction of the dataset. NeuralFP shares the audio files since they come from the FMA dataset [38]. It was trained on 10k FMA songs and tested on a larger set of 100k songs ($\approx$ 8,000 hours).

3. BAF DATASET

This section describes the characteristics of BAF: Broadcast Audio Fingerprinting dataset. It is the only available dataset designed for broadcast monitoring. BAF contains TV recordings, reference tracks, and annotations done by 6 different annotators that cross-annotated matching queries and references. It is a self-contained dataset available upon user's access request. Open for non-commercial, research-only, with no adaptations or derivative works allowed and proper attribution. It must not be used for music generation or music synthesis research. Towards addressing ethical and sustainability concerns, we distribute a datasheet using the format proposed by Gebru et al. [39] with practical and detailed information about the dataset. Audio files, annotations, and the dataset datasheet are hosted in Zenodo [1] while the baseline code and evaluation scripts are in Github [2].

3.1 Methodology

The reference set contains 2,000 production music tracks (74 hours of audio) obtained directly from the Epidemic Sound [9] private catalog, described in Section 3.3.2. The queries are initially derived from TV stream monitoring on 478 TV channels from 43 countries, for a 2.5 months period at stereo maximum quality. We extract the audio with FFMPEG and we split the large audio files into 1-minute length queries.

As an automatic pre-annotation stage, we match queries with references relying on a proprietary stereo matching algorithm developed by BMAT. The algorithm is non-scalable and it relies on spectral peaks matching using stereo signals. It has been tailored to avoid false negatives, disregarding the presence of false positives and low computational efficiency. We then select all query segments that have at least one pre-annotation match. In addition, we discard queries matching more than 3 unique references since most of them contain false positives, which would be manually deleted during the later annotation process. Then, we shuffle all queries and select 3,425 of them, corresponding to 57 hours of TV broadcast audio from 203 TV channels across 23 countries. Finally, we convert all audios to publishing format: 8kHz mono, WAV `pcm_s16le`, a common specification in AFP [2, 14, 15, 34].

3.2 Annotation criteria

BAF has been annotated by six different annotators in a controlled environment. We built an in-house annotation web app in Django, secured by user credentials, in order to facilitate the annotation to remote users. The app displays all segments resulting from the automatic pre-annotation stage. Annotators were instructed to listen to the query and reference pairs, filter out false positives, and adjust the start and end times of the true positives with deciseconds precision. Queries with slight alterations with respect to the reference have also been annotated as true positives, such as versions with some missing stem (e.g. instrumental vs vocal), or 'edit' versions.

We ensured that each segment was annotated by three different annotators (sets of annotators created by random combination). We then created the cross-annotations with 3 different levels of agreement: *single, majority, unanimity*. These cross-annotations are the result of splitting annotations into segments and merging overlapping segments, assigning a tag depending on how many annotators marked a match in that time interval (1, 2, and all 3, respectively). Out of the 57 hours of queries, over 37 hours were marked as true positive by at least 1 annotator.

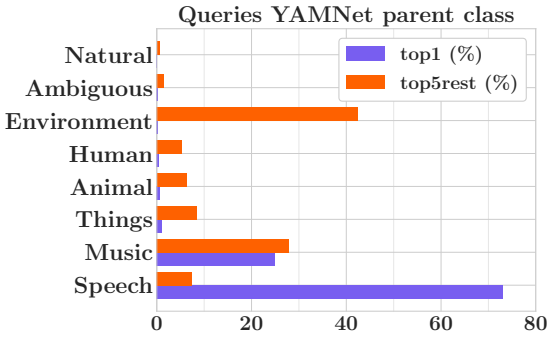


Figure 1. YAMNet classification of BAF queries. YAMNet uses a sliding window of length 0.96s and stride of 0.48s to generate one prediction for each step. Percentages are relative to each top classification.

3.3 Analysis

3.3.1 Queries

We have used the YAMNet sound event classifier [40] to study the most common sounds of BAF queries. YAMNet is a pretrained deep network that predicts 521 audio event classes based on the AudioSet-YouTube corpus [41]. Audioset ontology follows a tree structure so all classes are gathered under 7 different parent classes [42], from that, we extract *Speech* from *Human* as a separate class to better evaluate its presence. Figure 1 reflects that YAMNet's most predominant class is *Speech*, with 73% of the output predictions while only nearly 25% of them correspond to *Music*. Regarding the *top5rest* classes, *Environment* and *Music* are the most predominant, which means that noises and background music are common in the broadcast.

To obtain the YAMNet distributions we first run YAMNet for all queries and then select the outputs corresponding to the annotated segments. After that, we average the scores using a moving average window of length 2 to soften noisy scores. Lastly, we extract the top1 and top5 distributions and translate all labels to the corresponding parent class, following Audioset ontology. For each window, we remove from the top5 the parent class that matches that window's top1 parent class so with this, only the classes that are detected in the background remain. We name this distribution top5rest.

[1] `https://doi.org/10.5281/zenodo.6868083`
[2] `https://github.com/guillemcortes/baf-dataset`

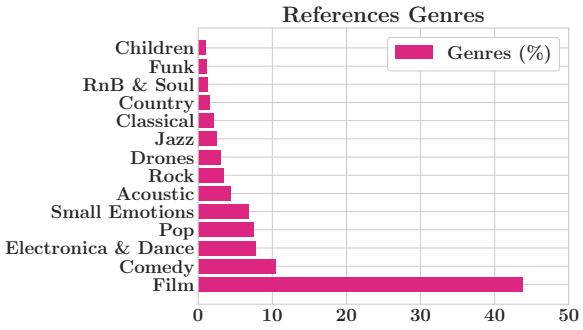


Figure 2. BAF references music genres. Only genres representing more than 1% are listed, the remainder represents 3% of the dataset.

In addition YAMNet predictions we used the MIREX 2019 Music Detection winner model [43, 44] to analyze BAF queries. The system is based on computing the Relative Music Loudness distribution [45] and classifies as *Foreground Music* only 18.07% of the total cross-annotated segments with the tag *unanimity* (segments all annotators agreed that there's music). Verifying the high presence of background music.

3.3.2 References

The BAF reference set is a selection of production music tracks from Epidemic Sound's private catalog of 35,000+ human-annotated tracks. In order to give valuable insights about instrumentalization, BPM, or genre, we have analyzed the tags and found that 7% of the selected tracks contain vocal elements like singing, while the remaining 93% are instrument-only versions. Figure 2 shows how the majority of the references are categorized as *Film music*, and cover styles/moods like Suspense, Drama, Build, Pulses, Small Emotions, or Solo Piano. Common keywords or tags found in this set of tracks include: comedic, piano, strings, driving, tension, corporate, guitar, and documentary. The references BPMs follow a Normal Distribution $\mathcal{N}(\mu, \sigma^2)$ with mean $\mu = 109$ and standard deviation $\sigma = 33$.

3.3.3 Matches / Annotations

Table 2 shows that 90.41% of the total annotated time (133,846 seconds) has been agreed by unanimity. Most of the differences between annotators come from divergences in start and end matching timestamps partially due to the difficulty to tell when a song starts or ends in a stream, especially with background music.

Additionally, to evaluate the reliability of annotators' agreement we compute the Fleiss' Kappa [46] indicator, a statistical measure of inter-rater reliability. It needs fixed-length elements to categorize them so we computed the Fleiss' Kappa indicator using 0.05 seconds length annotations. We obtained a factor of 0.9364 that can be interpreted as an almost perfect agreement [47].

3.4 Limitations

The size of the reference set is not large enough to mimic real production environments, where recordings are expected to be analyzed against tens of millions of tracks [25]. For future work, in order to study thoroughly the impact of False Positives (FP), a set of additional *noise* tracks should be added to the reference set. For this, public datasets mentioned in Section §2 could be used.

4. BENCHMARK

We benchmark the following available AFP algorithms: Audfprint [10], Panako [11, 12], Olaf [13], and NeuralFP [14] to study the performance of AFP in the broadcast monitoring use case. Additionally, we also propose a simple baseline that gives context to the results.

Audfprint is based on Shazam's algorithm [2]. It uses the locations of pairs of spectrogram peaks (local maxima points) as robust features for matching. **Panako** extends Shazam fingerprint by saving triplets of local maxima points in Constant-Q non-stationary Gabor transform. It uses time ratios to form a time-scale invariant fingerprint component. These components are hashed alongside a coarse value of the frequency position of the triplet, making it robust to time-scale and pitch modifications. **Olaf** is a lightweight AFP algorithm able to run in embedded systems. In the benchmarked version, it uses absolute exact frequencies and timestamps in the hash (like Shazam [2]) of triplets of peaks (like Panako [11]). It is not robust to pitch shifting or time distortions. **NeuralFP** is based on deep neural networks and created for high-specific audio retrieval using contrastive learning. It creates pairs of data applying distortions to short audio snippets so each batch of training data consists of randomly selected original samples and their augmented replicas. Then, it maximizes the inner product between pairs.

All algorithms except NeuralFP ran as they are published. NeuralFP, though, required changes in the indexing and matching modules to match the broadcast monitoring use case. The indexer now stores indexes on disk rather than on a memory map, and the matcher integrates the Maximum Inner Product Search used by the authors into PeakFP matcher pipeline, to be able to give start and end times of each match. The implementation [3] has been

[3] https://github.com/guillemcortes/neural-audio-fp

Class	%
single	3.69%
majority	5.90%
unanimity	90.41%

Table 2. Annotators agreement in percentage of annotation time length. *single* correspond to intervals where only 1 of the 3 annotators marked a match. In *majority* 2/3 annotators agreed while in *unanimity* there's full agreement.

validated by NeuralFP authors. Towards a fair comparison between systems, all algorithms return top1 matches. Audfprint and Olaf use the default configurations, Panako parameters are adjusted for 8kHz input signal and NeuralFP needs extra parameters for the custom matcher pipeline. Additionally, we study the impact of fingerprint density by increasing Audfprint (x2) and Panako (x1.5) peak density and also test two additional NeuralFP models *spcm1510* and *spc3000* that were trained with different levels of speech intensity [-15, 10] dB and [0, 10] dB, respectively. All configuration parameters used in this publication have been discussed with the authors of each respective algorithm and are available in the publication git repository.

Apart from the algorithms mentioned above, there are some others that we would have liked to benchmark, but no official public implementation of them was found. That is Fenet's et al. CQT approach [20], Google's Now Playing [26] lightweight, neural network-based, continuous monitoring system, and also Son's et al. FFMAP-based algorithm [27,48]. For QuadFP [23,49] and Waveprint [17] we tried to run third-party implementations but without success. We also plan to include Chromaprint [50], a public AFP implementation based on Ke et al. computer vision approach for music identification [51], and other algorithms to the benchmark.

4.1 Baseline: PeakFP

Available AFP systems aim at obtaining the best algorithm in terms of robustness, scalability, efficiency, etc. However, many existing systems are distributed as closed software packages or embedded into complicated frameworks which are difficult to adapt. Also, the literature lacks a simple and easy-to-use baseline that may be used as a starting point in AFP research. We address these issues by introducing PeakFP, a simple, open, non-data-driven, non-scalable, AFP algorithm based on spectral peak matching that it has been designed to be as simple as possible and not optimized for scalability, but at the same time, useful for detecting background music. While algorithms commonly use pairs or triplets of spectral peaks, PeakFP uses single-peak matching because pairs of spectral peaks are more prone to break when the SNR of the music signal is low due to music peaks being masked by other sounds. As a consequence, PeakFP is ineffective in front of pitch-shifting or time-scaling distortions.

PeakFP is divided into three modules: extractor, indexer, and matcher. They are designed to work independently following a simple pipeline we detail below. Note that the code of our implementation is available online (see Section 3). The extraction process involves finding peaks in a monaural audio magnitude spectrogram using a 2D max filter. Then, all the peaks (time-frequency tuples) are sorted by the time frame index and saved in a serialized binary file. The reference signature files are used to generate an inverted index on the peak frequency values. For a given frequency, the index contains the list of all occurrences of that frequency in every reference. The hash space comprises all the possible frequency values. This small hash space translates to a high quantity of hash matches that yield a high number of comparisons in the matching step. The matcher splits the peaks of the query recordings using a sliding temporal window defined by window length and hop size. Then, for all windows, it counts the common peaks for a specific query-reference time alignment similarly to Shazam [2]. After that, all matches go through a postprocessing stage in order to consolidate and resolve overlapping matches.



Figure 3. Proposed metric Match ratio. It defines the ratio between the number of identifications with respect to the number of annotations in the groundtruth.

4.2 Evaluation Metrics

A variety of metrics are used in the AFP literature for benchmarking and comparing algorithms, most of them based on classifying predictions into false positives/negatives or true labels [52], and using metrics derived from information retrieval such as True Positive Rate [11] Precision, Recall, and Specificity [2,23]. Other papers use Top-1 Hit Rate [14] or TRECVID's proposed evaluation metric Normalized Detection Cost Rate (NDCR) [33–35].

In broadcast monitoring, it is typically required to provide exact start and end matching timestamps for each reference identification. For this reason, we propose to use the percentage of identified seconds alongside to match classification into Precision, Recall, and F1-score.

Some algorithms are prone to give short overly-split matches, while others tend to generate longer matches that include gaps without annotations. Towards quantifying this we introduce a new metric *Match Ratio*, defined in equation 1 and depicted in Figure 3, that represents the ratio between the total correct identified segments (TP matches), and the total unique annotations identified, groundtruth (GT) segments.

$$\text{Match Ratio} = \frac{\#\text{ TP segments ID}}{\#\text{ TP segments GT}} \quad (1)$$

A Match Ratio value bigger than 1 means that some of the identifications belong to the same annotation. Conversely, a value lower than 1 indicates that the algorithm generates a single identification for a segment in which there is more than one unique identification according to the groundtruth. The value for an algorithm that perfectly matches the annotations is 1.

Algorithm	Match Ratio	# matches		seconds identified		
		Precision	GT Recall	Precision	Recall	F1-score
PeakFP (baseline)	1.64	.96	**.72**	.96	**.32**	**.47**
Panako2.0	1.85	**.98**	.21	**.98**	.06	.12
Panako2.0 (x1.5)	2.12	.70	.41	.69	.15	.25
Olaf	1.95	**.98**	.14	**.98**	.06	.11
NeuralFP	1.39	.22	.23	.37	.10	.15
NeuralFP-spcm1510	1.56	.23	.45	.38	.22	.28
NeuralFP-spc3000	1.40	.69	.31	.83	.13	.22
Audfprint	N/A*	.76	.05	.86	.02	.04
Audfprint (x2)	N/A*	.71	.10	.81	.04	.08

Table 3. Benchmark results on *unanimity* annotations. *Audfprint reports 1 match per query by default.

4.3 Results

Table 3 summarizes the performance of all benchmarked algorithms on *unanimity* annotations. All systems increase their Precision when considering the identified seconds because the False Positives identifications are shorter than the True Positives. At the same time, Recall decreases because the identifications are partial and do not cover the full annotation groundtruth. Hence the importance of studying also the performance in terms of identified seconds.

All algorithms except Audfprint obtain a Match Ratio > 1. This is caused because algorithms tend to detect small excerpts of the music (parts where the music SNR is higher) resulting in more than one identification per annotation. Audfprint only returns one identification per query by default, so its Match Ratio will always be 1 with this configuration. This also means that if a query has more than one annotation, Audfprint can't identify all of them.

The good Precision results (PeakFP, Panako, Olaf obtain over 0.96) should be analyzed taking into account that BAF is not challenging to False Positives since the reference set is limited to 2,000 references. The low Recall values show that there are a lot of identifications missed (False Negatives), manifesting that algorithms do not work well with background music or broadcast distortions. Increasing fingerprint density improves the F1-score in seconds identified: it helps to boost the Recall in both Panako (x1.5) and Audfprint (x2) but at expense of Precision.

Additionally, to frame the computational cost of each algorithm, we have benchmarked their extraction, indexing, and matching times as well as index size. Table 4 shows that Olaf is the fastest benchmarked system. On the other hand, PeakFP is the slowest even though the extraction process is quick due to its simplicity, but the reduced hash space yields a high collision of hashes that slows down the matching. NeuralFP extraction process takes a lot of time compared to others, mainly due to the deep neural network model complexity. This process can be sped up by running it on a GPU, where deep learning models operate more efficiently. For both Panako and Audfprint, the matching step takes x2.5 times longer than the extraction.

Algorithm	Extraction & Indexing	Matching	Index size
Olaf	53m	3h 30m	349 MB
Panako 2.0	2h 17m	5h 24m	273 MB
NeuralFP	49h 34m	9h 30m	37 MB
Audfprint	9h 50m	23h 01m	19 MB
PeakFP	50m	98h 39m	160 MB

Table 4. Computational cost benchmark.

Regarding the index sizes, Audfprint and NeuralFP generate the smallest indexes (19MB and 37 MB) to represent a set of 2,000 references (74 hours of audio). Olaf generates the biggest index with almost 350 MB, 18 times the size of the Audfprint index. Olaf would take around 1.75 TB for an industrial database of 10 million references while Audfprint would take 95 GB.

Experiments have been run in a reproduceable, isolated environment. All audios have been loaded on the RAM disk and ran on a 98GB RAM server with two 16-cores CPUs at 2.60GHz. We have executed each algorithm with multiprocessing (when it was possible) and cleared the cache before each run. The results in Table 4 are normalized to one single thread.

5. CONCLUSIONS AND FUTURE WORK

We present a new dataset for broadcast monitoring with 57 hours of TV broadcast recordings, 74 hours of production music, and over 37 hours of human cross-annotations. All audios are monaural sampled at 8kHz and more than 80% of the annotated music is in the background. The Benchmark of state-of-the-art public algorithms shows that AFP for broadcast monitoring with a high presence of background music is yet to be solved. For this task, in addition to other metrics used in the literature, we propose using the Match Ratio and analyzing Precision, Recall, and F1-score, especially in terms of seconds identified. We also provide a simple AFP baseline named PeakFP.

In future experiments, we plan to add noise tracks to the references set to study the evolution of False Positives and the scalability of each algorithm. We will get more insight into their behavior for different levels of background music, and we will benchmark more AFP algorithms.

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7. REFERENCES

[1] P. Cano, E. Batlle, E. Gómez, L. de C. T. Gomes, and M. Bonnet, "Audio fingerprinting: Concepts and applications," in *Computational Intelligence for Modelling and Prediction*, ser. Studies in Computational Intelligence, S. K. Halgamuge and L. Wang, Eds. Springer, 2005, vol. 2, pp. 233–245. [Online]. Available: https://doi.org/10.1007/10966518_17

[2] A. Wang, "An industrial strength audio search algorithm," in *ISMIR 2003, 4th International Conference on Music Information Retrieval, Baltimore, Maryland, USA, October 27-30, 2003, Proceedings*, 2003, pp. 7–13.

[3] J. R. Cerquides, "A real time audio fingerprinting system for advertisement tracking and reporting in fm radio," in *2007 17th International Conference Radioelektronika*. IEEE, 2007, pp. 1–4.

[4] E. Gomez, P. Cano, L. Gomes, E. Batlle, and M. Bonnet, "Mixed watermarking-fingerprinting approach for integrity verification of audio recordings," in *Proceedings of the International Telecommunications Symposium*, 2002.

[5] C. J. C. Burges, D. Plastina, J. C. Platt, E. Renshaw, and H. S. Malvar, "Using audio fingerprinting for duplicate detection and thumbnail generation," in *2005 IEEE International Conference on Acoustics, Speech, and Signal Processing, ICASSP '05, Philadelphia, Pennsylvania, USA, March 18-23, 2005*. IEEE, 2005, pp. 9–12. [Online]. Available: https://doi.org/10.1109/ICASSP.2005.1415633

[6] IFPI, "Global music report 2022," https://www.ifpi.org/wp-content/uploads/2022/04/IFPI_Global_Music_Report_2022-State_of_the_Industry.pdf, 4 2022, [Accessed August 2022].

[7] B. Meléndez Catalán *et al.*, "Relative music loudness estimation in tv broadcast audio using deep learning: an industrial perspective," Ph.D. dissertation, Universitat Pompeu Fabra, 2021.

[8] A. G. Piotrowska, "Analyzing music in tv shows: some methodological considerations," *The science of television*, no. 14.3, pp. 10–27, 2018.

[9] "Epidemic sound," https://www.epidemicsound.com/, 2009, [Accessed August 2022].

[10] D. Ellis, "The 2014 labrosa audio fingerprint system," in *Proceedings of the 15th International Society for Music Information Retrieval Conference, ISMIR 2014, Taipei, Taiwan, TW, October 27-31, 2014*, 2014.

[11] J. Six and M. Leman, "Panako - A scalable acoustic fingerprinting system handling time-scale and pitch modification," in *Proceedings of the 15th International Society for Music Information Retrieval Conference, ISMIR 2014, Taipei, Taiwan, October 27-31, 2014*, H. Wang, Y. Yang, and J. H. Lee, Eds., 2014, pp. 259–264. [Online]. Available: http://www.terasoft.com.tw/conf/ismir2014/proceedings/T048_122_Paper.pdf

[12] J. Six, "Panako 2.0-updates for an acoustic fingerprinting system," in *Demo / late-breaking abstracts of 22st International Society for Music Information Retrieval Conference, ISMIR 2021, Online, November 8-12, 2021*, 2021.

[13] ——, "Olaf: Overly lightweight acoustic fingerprinting," in *Demo / late-breaking abstracts of 21st International Society for Music Information Retrieval Conference, ISMIR 2020, Montréal, Canada, CA, October 11-16, 2020*, 2020.

[14] S. Chang, D. Lee, J. Park, H. Lim, K. Lee, K. Ko, and Y. Han, "Neural audio fingerprint for high-specific audio retrieval based on contrastive learning," in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2021, Toronto, ON, Canada, June 6-11, 2021*. IEEE, 2021, pp. 3025–3029. [Online]. Available: https://doi.org/10.1109/ICASSP39728.2021.9414337

[15] J. Haitsma and T. Kalker, "A highly robust audio fingerprinting system," in *ISMIR 2002, 3rd International Conference on Music Information Retrieval, Paris, France, October 13-17, 2002, Proceedings*, 2002, pp. 107–115. [Online]. Available: http://ismir2002.ismir.net/proceedings/02-FP04-2.pdf

[16] M. A. Bartsch and G. H. Wakefield, "Audio thumbnailing of popular music using chroma-based representations," *IEEE Trans. Multim.*, vol. 7, no. 1, pp. 96–104, 2005. [Online]. Available: https://doi.org/10.1109/TMM.2004.840597

[17] S. Baluja and M. Covell, "Waveprint: Efficient wavelet-based audio fingerprinting," *Pattern Recognit.*, vol. 41, no. 11, pp. 3467–3480, 2008. [Online]. Available: https://doi.org/10.1016/j.patcog.2008.05.006

[18] C. Bellettini and G. Mazzini, "Reliable automatic recognition for pitch-shifted audio," in *Proceedings of the 17th International Conference on Computer Communications and Networks, IEEE ICCCN 2008, St. Thomas, U.S. Virgin Islands, August 3-7, 2008*.

IEEE, 2008, pp. 838–843. [Online]. Available: https://doi.org/10.1109/ICCCN.2008.ECP.157

[19] ——, "A framework for robust audio fingerprinting," *J. Commun.*, vol. 5, no. 5, pp. 409–424, 2010. [Online]. Available: https://doi.org/10.4304/jcm.5.5.409-424

[20] S. Fenet, G. Richard, and Y. Grenier, "A scalable audio fingerprint method with robustness to pitch-shifting," in *Proceedings of the 12th International Society for Music Information Retrieval Conference, ISMIR 2011, Miami, Florida, USA, October 24-28, 2011*, A. Klapuri and C. Leider, Eds. University of Miami, 2011, pp. 121–126. [Online]. Available: http://ismir2011.ismir.net/papers/PS1-14.pdf

[21] M. Malekesmaeili and R. K. Ward, "A local fingerprinting approach for audio copy detection," *Signal Process.*, vol. 98, pp. 308–321, 2014. [Online]. Available: https://doi.org/10.1016/j.sigpro.2013.11.023

[22] X. Zhang, B. Zhu, L. Li, W. Li, X. Li, W. Wang, P. Lu, and W. Zhang, "Sift-based local spectrogram image descriptor: a novel feature for robust music identification," *EURASIP J. Audio Speech Music. Process.*, vol. 2015, p. 6, 2015. [Online]. Available: https://doi.org/10.1186/s13636-015-0050-0

[23] R. Sonnleitner and G. Widmer, "Robust quad-based audio fingerprinting," *IEEE ACM Trans. Audio Speech Lang. Process.*, vol. 24, no. 3, pp. 409–421, 2016. [Online]. Available: https://doi.org/10.1109/TASLP.2015.2509248

[24] R. Sonnleitner, A. Arzt, and G. Widmer, "Landmark-based audio fingerprinting for DJ mix monitoring," in *Proceedings of the 17th International Society for Music Information Retrieval Conference, ISMIR 2016, New York City, United States, August 7-11, 2016*, M. I. Mandel, J. Devaney, D. Turnbull, and G. Tzanetakis, Eds., 2016, pp. 185–191. [Online]. Available: https://wp.nyu.edu/ismir2016/wp-content/uploads/sites/2294/2016/07/187_Paper.pdf

[25] T. Walther and M. D. Gould, "Audio identification method," Worldwide Patent WO/2016/189 307, 2016.

[26] B. A. y Arcas, B. Gfeller, R. Guo, K. Kilgour, S. Kumar, J. Lyon, J. Odell, M. Ritter, D. Roblek, M. Sharifi, and M. Velimirovic, "Now playing: Continuous low-power music recognition," *CoRR*, vol. abs/1711.10958, 2017. [Online]. Available: http://arxiv.org/abs/1711.10958

[27] H. Son, S. Byun, and S. Lee, "A robust audio fingerprinting using a new hashing method," *IEEE Access*, vol. 8, pp. 172 343–172 351, 2020. [Online]. Available: https://doi.org/10.1109/ACCESS.2020.3024951

[28] Z. Yu, X. Du, B. Zhu, and Z. Ma, "Contrastive unsupervised learning for audio fingerprinting," *CoRR*, vol. abs/2010.13540, 2020. [Online]. Available: https://arxiv.org/abs/2010.13540

[29] E. Law, K. West, M. I. Mandel, M. Bay, and J. S. Downie, "Evaluation of algorithms using games: The case of music tagging," in *Proceedings of the 10th International Society for Music Information Retrieval Conference, ISMIR 2009, Kobe International Conference Center, Kobe, Japan, October 26-30, 2009*, K. Hirata, G. Tzanetakis, and K. Yoshii, Eds. International Society for Music Information Retrieval, 2009, pp. 387–392. [Online]. Available: http://ismir2009.ismir.net/proceedings/OS5-5.pdf

[30] M. Ramona and G. Peeters, "Audio identification based on spectral modeling of bark-bands energy and synchronization through onset detection," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing, ICASSP 2011, May 22-27, 2011, Prague Congress Center, Prague, Czech Republic*. IEEE, 2011, pp. 477–480. [Online]. Available: https://doi.org/10.1109/ICASSP.2011.5946444

[31] A. F. Smeaton, P. Over, and W. Kraaij, "Evaluation campaigns and trecvid," in *Proceedings of the 8th ACM SIGMM International Workshop on Multimedia Information Retrieval, MIR 2006, October 26-27, 2006, Santa Barbara, California, USA*, J. Z. Wang, N. Boujemaa, and Y. Chen, Eds. ACM, 2006, pp. 321–330. [Online]. Available: https://doi.org/10.1145/1178677.1178722

[32] G. Awad, P. Over, and W. Kraaij, "Content-based video copy detection benchmarking at TRECVID," *ACM Trans. Inf. Syst.*, vol. 32, no. 3, pp. 14:1–14:40, 2014. [Online]. Available: https://doi.org/10.1145/2629531

[33] X. Anguera, A. Garzon, and T. Adamek, "MASK: robust local features for audio fingerprinting," in *Proceedings of the 2012 IEEE International Conference on Multimedia and Expo, ICME 2012, Melbourne, Australia, July 9-13, 2012*. IEEE Computer Society, 2012, pp. 455–460. [Online]. Available: https://doi.org/10.1109/ICME.2012.137

[34] C. Ouali, P. Dumouchel, and V. Gupta, "A robust audio fingerprinting method for content-based copy detection," in *12th International Workshop on Content-Based Multimedia Indexing, CBMI 2014, Klagenfurt, Austria, June 18-20, 2014*. IEEE, 2014, pp. 1–6. [Online]. Available: https://doi.org/10.1109/CBMI.2014.6849814

[35] Z. J. Guzman-Zavaleta, C. F. Uribe, A. Menendez-Ortiz, and J. J. Garcia-Hernandez, "A robust audio fingerprinting method using spectrograms saliency maps," in *9th International Conference for Internet Technology and Secured Transactions, ICITST 2014, London, United Kingdom, December 8-10, 2014*. IEEE, 2014, pp. 47–52. [Online]. Available: https://doi.org/10.1109/ICITST.2014.7038773

[36] NIST-TRECVID, "Trecvid 2011 - content-based copy detection task," https://www-nlpir.nist.gov/projects/

tv2011/index.html#ccd, 2010, [Accessed August 2022].

[37] J. D. of Computational Perception, "Audio fingerprinting data sets," http://www.cp.jku.at/datasets/fingerprinting, 4 2017, [Accessed August 2022].

[38] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, "FMA: A dataset for music analysis," in *Proceedings of the 18th International Society for Music Information Retrieval Conference, ISMIR 2017, Suzhou, China, October 23-27, 2017*, S. J. Cunningham, Z. Duan, X. Hu, and D. Turnbull, Eds., 2017, pp. 316–323. [Online]. Available: https://ismir2017.smcnus.org/wp-content/uploads/2017/10/75_Paper.pdf

[39] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. M. Wallach, H. D. III, and K. Crawford, "Datasheets for datasets," *Commun. ACM*, vol. 64, no. 12, pp. 86–92, 2021. [Online]. Available: https://doi.org/10.1145/3458723

[40] Tensorflow, "Sound classification with yamnet," https://www.tensorflow.org/hub/tutorials/yamnet, 2022, [Accessed August 2022].

[41] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio set: An ontology and human-labeled dataset for audio events," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2017, New Orleans, LA, USA, March 5-9, 2017*. IEEE, 2017, pp. 776–780. [Online]. Available: https://doi.org/10.1109/ICASSP.2017.7952261

[42] J. F. Gemmeke, D. P. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audioset ontology," https://research.google.com/audioset/ontology/index.html, 2017, [Accessed August 2022].

[43] MIREX, "2019: Music detection results," https://www.music-ir.org/mirex/wiki/2019:Music_Detection_Results, 2019, [Accessed August 2022].

[44] B. Meléndez-Catalán, "Relative music loudness estimation using temporal convolutional networks and a cnn feature extraction front-end," in *Proceedings of the 23rd International Conference on Digital Audio Effects (DAFx-20)*, vol. 5, 2020, pp. 273–280.

[45] B. Meléndez-Catalán, E. Molina, and E. Gómez, "Open broadcast media audio from TV: A dataset of TV broadcast audio with relative music loudness annotations," *Trans. Int. Soc. Music. Inf. Retr.*, vol. 2, no. 1, pp. 43–51, 2019. [Online]. Available: https://doi.org/10.5334/tismir.29

[46] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychological bulletin*, vol. 76, no. 5, p. 378, 1971.

[47] A. J. Viera, J. M. Garrett *et al.*, "Understanding interobserver agreement: the kappa statistic," *Fam med*, vol. 37, no. 5, pp. 360–363, 2005.

[48] H. Son, S. W. Byun, and S. Lee, "Illegal audio copy detection using fundamental frequency map," in *Proceedings of the 16th International Joint Conference on e-Business and Telecommunications, ICETE 2019 - Volume 1: DCNET, ICE-B, OPTICS, SIGMAP and WINSYS, Prague, Czech Republic, July 26-28, 2019*, M. S. Obaidat, C. Callegari, M. van Sinderen, P. Novais, P. G. Sarigiannidis, S. Battiato, Á. S. S. de León, P. Lorenz, and F. Davoli, Eds. SciTePress, 2019, pp. 356–361. [Online]. Available: https://doi.org/10.5220/0008113403500355

[49] R. Sonnleitner and G. Widmer, "Quad-based audio fingerprinting robust to time and frequency scaling," in *Proceedings of the 17th International Conference on Digital Audio Effects, DAFx-14, Erlangen, Germany, September 1-5, 2014*, S. Disch, J. Herre, R. Rabenstein, B. Edler, M. Müller, and S. Turowski, Eds., 2014, pp. 173–180. [Online]. Available: http://www.dafx14.fau.de/papers/dafx14_reinhard_sonnleitner_quad_based_audio_fingerpr.pdf

[50] L. Lalinský, "Chromaprint," https://acoustid.org/chromaprint, [Accessed August 2022].

[51] Y. Ke, D. Hoiem, and R. Sukthankar, "Computer vision for music identification," in *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2005), 20-26 June 2005, San Diego, CA, USA*. IEEE Computer Society, 2005, pp. 597–604. [Online]. Available: https://doi.org/10.1109/CVPR.2005.105

[52] M. Ramona and G. Peeters, "Audioprint: An efficient audio fingerprint system based on a novel cost-less synchronization scheme," in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2013, Vancouver, BC, Canada, May 26-31, 2013*. IEEE, 2013, pp. 818–822. [Online]. Available: https://doi.org/10.1109/ICASSP.2013.6637762